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CreativeForge AI – Full Project Documentation

Tool Name and Link to access it: CreativeForge AI (<https://roaring-ink-forge-flow.base44.app>)

Type of Implementation: No-code AI workflow built using Base44

Overview

The CreativeForge AI generator is a no-code AI system that produces creative writing outputs (stories, poems, and dialogue scenes) based on user-selected inputs. The system uses structured prompt templates combined with dynamic variable injection to generate personalized content in real time.

User Input Interface

- Content Type (Short Story / Poem / Dialogue)
- Genre (Fantasy, Romance, Horror, Sci-Fi, Motivational)
- Tone (Emotional, Dark, Funny, Inspirational)
- Length (Short / Medium)
- Audience (Kids, Teens, Adults)
- Keywords (Optional, max 5)

Workflow Process

- User enters input parameters
- System validates inputs (required fields + keyword limits)
- Prompt template is selected based on content type
- User inputs are injected into the prompt at generation time
- AI model generates creative output
- Output filtering removes unsafe or irrelevant content
- Final content is displayed to the user
- Performance metrics are recorded (time + tokens)
- User can save or export the result

Core Features

- Dynamic prompt generation
- Structured storytelling formats
- Audience-based adaptation

- Input validation system
- Output safety filtering
- Error handling for API failures
- Performance tracking (generation time + token usage)
- Save output to library and export functionality in TXT and PDG form



Prompt Library

1. Short Story Generation Template

Purpose: Generate structured short stories with clear narrative flow. Inputs: tone, genre, length, audience, keywords

Template:

Write a {tone} {genre} short story of {length} for a {audience} audience. Include keywords: {keywords}

Format:

- Title
- Introduction
- Conflict
- Resolution

Variation: Dark/emotional version focuses on psychological depth.

Usage: Short Story selection

2. Poem Generation Template

Purpose: Generate emotionally expressive poems. Inputs:

tone, genre, length, audience, keywords **Template:**

Write a {tone} {genre} poem of {length} for a {audience} audience. Include keywords: {keywords}

Format:

- Title
- Poem lines

Variation: Inspirational version focuses on hope and growth.

3. Dialogue Scene Template

Purpose: Generate cinematic dialogue scenes. Inputs: tone,

genre, length, audience, keywords **Template:**

Write a {tone} {genre} dialogue scene of {length}.

Format:

- Title
- Scene 1
- Scene 2
- Final Scene

Variation: Psychological tension version.

4. Motivational Story Template

Purpose: Inspirational storytelling.

Inputs: tone, genre, length, audience, keywords

Template:

Write a motivational {genre} story of {length}.

Structure:

- Introduction
- Conflict
- Resolution

Variation: School motivation version.

5. Sci-Fi Adventure Template

Purpose: Futuristic storytelling.

Inputs: tone, genre, length, audience, keywords

Template:

Write a sci-fi adventure story of {length}.

Structure:

- Introduction
- Conflict
- Resolution

Variation: AI future version.

Comparison Matrix

Prompt Version	Improvement Added	Quality Results
V1	Basic Prompt	Generic outputs
V2	Added tone control	Better emotional consistency
V3	Added Audience conditioning	More personalized output
V4	Added formatting structure	Clear and organized content
V5	Added filtering and constraints	Highest-quality outputs

Sample Outputs

1. Short Story for adults (Fantasy, Emotional)

The Mirror at the Threshold of Everything

Introduction

She had crossed a thousand thresholds in her long life — doorways of stone, of shadow, of grief — but none had stopped her breath the way this one did. The mirror stood at the forest's edge, framed in roots that had grown around it over centuries, as though the earth itself had claimed it. Its surface did not show the trees behind her, nor the amber light of the failing sun. It showed a girl she had forgotten she had ever been. Her fingers rose before her mind could protest. The glass was warm.

Conflict

"You left me here," the reflection whispered — and it was her voice, younger and unbroken. The words landed like stones in still water, each ripple reaching further than the last.

She had built her entire existence on forward motion. She had told herself that what she left behind was weight, not self. But now the forgotten version of her stood with wet cheeks and open hands, and she could not look away. The forest pressed closer. The roots grew. The mirror began to cloud from the edges inward, slowly erasing the girl — erasing the last honest version of her.

Stay everything in her begged. Go, said the part of her that had survived this long.

Resolution

She pressed both palms flat against the glass. The warmth spread up her arms, into her chest, into the hollow place she had plastered over with purpose and distance. She did not step through. She did not need to.

"I remember you," she said aloud, and meant it for the first time.

The mirror cleared. The reflection smiled — not with relief, but with recognition. The roots loosened. The forest exhaled.

She turned back toward the path, carrying something she had not carried in decades: the full weight of herself.

2. Short Poem for adults (Horror, Dark)

What the Hollow Holds

*The wire runs beneath the floor,
thin as a whisper, tight as a vow —
you only feel it when you've stopped
believing in the walls.*

*The descent is never sudden.
It is a slow forgetting:
first the daylight at the edge of rooms,
then the names of things you loved,
then the shape of your own hands
in the dark.*

*Something lives inside the hollow.
It does not hunger — it waits.
It has learned that patience
is a sharper instrument than teeth.*

*When you finally press your ear
to the place where the sound
used to be —*

*you will hear only your own breathing,
and beneath it,
something breathing back.*

3. Funny Dialogue for Teens(Sci-Fitone, keywords: wormhole, lunch, homework)

The Wormhole Ate My Homework (And My Lunch)

Scene 1: The School Cafeteria — Fifteen Minutes After the Incident

MAYA: Okay. I need you to explain this slowly and without using the word 'quantum.

DECKER: The wormhole opened exactly where I put my tray down.

MAYA: So your lunch is... where?

DECKER: Best case? Andromeda. Worst case? Tuesday.

MAYA: Tuesday.

DECKER: Time is less linear than they told us in Year 8, Maya.

Scene 2: The Library — Decker Attempts to Recover His Homework

MAYA You cannot submit 'a wormhole ate it' to Mr. Finch. He's already suspicious of you since the helium incident.

DECKER: That was a controlled experiment.

MAYA: You floated into the ceiling.

DECKER: I observed the ceiling at close range.

MAYA: (typing furiously) If I can recalibrate the aperture using the school's Wi-Fi router—

DECKER: You're going to use the school Wi-Fi to navigate a spacetime anomaly?

MAYA: It's 6G. It's basically the same thing.

4. Motivational and Inspirational Short story for Kids(keywords: pencil, cloud, brave)

The Brave Little Cloud and the Magic Pencil

Introduction

Pip was a small cloud who lived in a big, busy sky. All the other clouds made rain and thunder and great, roaring storms — but whenever Pip tried, only tiny drizzles came out. Pip carried a magic pencil everywhere, just in case.

Adventure

One summer morning, a girl named Rosie sat below with her sketchbook, sad because she could not think of anything to draw. Pip looked down, took a deep brave breath, and drew a rainbow across the whole sky using the magic pencil.

Happy Ending

Rosie looked up and gasped with delight. She filled her whole sketchbook with colours.

Pip floated home feeling ten times bigger than before.

Being brave, Pip learned, did not mean making the biggest storm. Sometimes it meant making the brightest rainbow.

5. Sci-Fi Inspirational Poems for Teens (keywords: signal, distance, light)

The Signal That Crossed Everything

*They said the distance was too great —
that light itself would lose the will
to travel that far, that long,
through the indifferent black.*

*They measured it in years
and called it impossible
the way adults call things impossible
when they are only afraid.*

But you sent the signal anyway.

*You aimed it at the dark
with the recklessness of someone
who still believes that effort matters,
that the universe is listening,
that somewhere beyond the noise
there is a frequency that answers.*

CreativeForge AI – Technical Documentation

Overview

CreativeForge AI was designed using a structured no-code AI application architecture focused on creative content generation, user safety, prompt engineering, and workflow automation.

Implementation Architecture

The system is composed of the following layers:

1. User Interface Layer which collects user inputs like, content, Type, Genre, Tone, Length, Audience, and Keywords.
2. Input Validation Layer, To ensure that all required fields are completed. Enforces keyword limits and prevents invalid or empty submissions.
3. Prompt Engineering Layer, Which dynamically constructs AI prompts using selected parameters. It routes inputs into predefined prompt templates and applies formatting rules based on content type.
4. AI Generation Layer sends structured prompts to the Base44 AI model to generate creative writing outputs.
5. Output Filtering Layer removes inappropriate or irrelevant content to ensure audience-appropriate responses
Applies safety moderation rules
6. Output Display Layer, Which presents generated content in structured format. Includes sections such as title, story, or poem formatting.
7. Utility Layer Perform tracking (time + tokens), Save functionality and Export functionality.

Configure Your Piece
Select your preferences and let AI craft your content.

Content Type

Short Story Poem Dialogue

Genre

Fantasy Romance Horror Sci-Fi Motivational

Tone

Emotional Dark Funny Inspirational

Please select a tone

Length

Short (~150 words) Medium (~350 words)

Please select a length

Audience

Kids Teens Adults

Please select an audience

Keywords (optional, max 5)

Type a keyword and press Enter...

Generate Content

Figure 1: CreativeForge AI main user interface

API Selection Rationale

Platform used is Base44 platform because it has:

- No-Code Integration, Eliminates the need for manual backend development and Enables fast AI workflow creation.
- Built-in AI Workflow Support as it Supports structured prompt execution and allows dynamic variable mapping.
- Scalability, It is suitable for expanding AI applications Supports multiple prompt templates and workflows.
- Efficiency, It reduces development time and provides fast prototyping capabilities.
- Built-in Safety Features which Supports moderation and filtering mechanisms also helps enforce safe content generation.

Prompt Engineering Methodology

The application uses a parameterized template-based prompt engineering architecture, where a fixed prompt scaffold is combined with run-time variable injection. Instead of generating free-form instructions per request, the system relies on predefined structural templates in which only controlled variables (genre, tone, audience, keywords, and length) are dynamically inserted.

This approach ensures consistency, reduces prompt uncertainty, and improves output reliability across different creative writing formats.

Key Techniques Applied:

1. Structure-First Prompt Design

Each prompt defines the exact output structure (using markdown formatting) before any creative instructions are given. This ensures the model adheres to consistent formatting rules such as headings, sections, and spacing.

2. Behavioral Audience Conditioning

Rather than using simple audience labels (e.g., “Kids”), the system applies explicit behavioral instructions such as using simple vocabulary.

This reduces interpretation variance and improves age-appropriate responses.

2. Output Isolation Directive

Each prompt includes strict output control instructions such as “Output only the formatted content.” This eliminates unwanted explanations, preambles, or meta-text, ensuring clean integration with the user interface.

4. Structural Anchoring Using Section Markers

Emoji-based section headers are used as visual anchors. These improve readability and help reinforce structured narrative segmentation within generated outputs.

5. Keyword Natural Integration Constraint

Keywords are injected with the instruction “incorporate naturally,” ensuring they are embedded contextually within the narrative instead of being forcefully inserted, preserving linguistic fluency.

6. Modular Content-Type Prompt Dispatch

Separate prompt templates are used for each content type (story, poem, dialogue). This modular design improves maintainability, scalability, and allows independent optimization of each creative format.

Performance Optimization Techniques used:

- **Prompt Reusability:**
Standardized templates reduce redundancy
Improves efficiency across multiple generations
 - **Token Optimization:**
Prompts are structured to avoid unnecessary text
Reduces API cost and response time
 - **Validation Before Execution:**
Prevents unnecessary API calls
Reduces system load
 - **Efficient Prompt Design:**
Clear instructions reduce AI confusion
Improves output accuracy
- Retry Logic (No-Code Implementation)

Limitation Management Strategies

The system includes strategies to handle technical and AI limitations:

- **The API Rate Limiting** controls request frequency and prevents system overload.
- **Output Inconsistency Handling:**
Structured prompts reduce variation issues and formatting rules enforce consistency.
- **Empty Output Handling:**
System detects blank responses also automatically prompts regeneration.
- **Input Error Prevention:**
Validation prevents invalid prompts and reduces system failures.

Conclusion

CreativeForge AI is a no-code AI-powered creative writing tool that generates short stories, poems, and dialogue scenes based on user-selected inputs such as tone, genre, audience, length, and keywords. The system uses structured prompt templates and dynamic variable injection to ensure consistent, high-quality, and personalized outputs.

The application also includes input validation, output filtering, error handling, performance tracking, and export functionality to improve reliability and user experience. Overall, the project demonstrates effective use of prompt engineering and no-code AI workflows to build a functional and user-friendly creative writing generator.